

Naveen Manikandan

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SUMMARY

MS Computer Science graduate from the University at Buffalo (May 2026) with 3 years of experience at TCS building Java microservices and enterprise data pipelines across SAP, Oracle, GCP and Salesforce. Experienced in backend development, data engineering and applied AI, including a vision-language model project that reduced hallucination rates by 82% without model retraining. Skilled in developing scalable backend applications, data pipelines, cloud-based systems and modern front-end development using React. Seeking Software Engineer, Data Engineer and AI Platform roles. Open to relocation across the U.S.

EDUCATION

State University of New York at Buffalo

January 2025 – May 2026

Master of Science in Computer Science and Engineering

Buffalo, NY

Coursework: Machine Learning, Deep Learning, Data Intensive Computing, Algorithms, Computer Security, Modern Network Concepts, Computer Architecture

SKILLS

Languages	Java, Python, JavaScript, SQL, Shell Scripting
Frontend	React.js, HTML5, CSS3, Responsive Web Design, Component-Based Development
Backend	Spring Boot, Hibernate (JPA), REST APIs, Microservices, Apache Kafka, OAuth 2.0, JUnit
Data Engineering	ETL Pipelines, Informatica, Data Integration, SQL Optimization, Pandas, NumPy
AI / ML	PyTorch, PySpark, TensorFlow, Scikit-learn, Hugging Face Transformers, BLIP, CLIP, Vision-Language Models, Computer Vision
Cloud & DevOps	Docker, Kubernetes, Azure DevOps (CI/CD), Jenkins, Git, GCP
Databases	MySQL, MS SQL, SAP HANA, Oracle

PROJECTS

Hallucination Grounding in Vision-Language Models | Python, PyTorch, BLIP, CLIP, Hugging Face

April 2026

- Designed a training-free caption selection framework using BLIP and CLIP to reduce hallucinated object mentions in generated captions without any model retraining.
- Scored 6 candidate captions per image through a hybrid function combining CLIP visual alignment with object-count and length penalties to select the most grounded output.
- Reduced sentence-level hallucination rate from 4.08% to 0.74% across 5,000 COCO val2017 images, an 82% relative reduction, after restricting evaluation to COCO where CLIP alignment bias on Flickr30k was identified.
- Ran a 7 configuration ablation study revealing CLIP alignment alone worsens hallucination, with the object-count penalty contributing most to the improvement.

Earthquake Risk Analysis | Python, PySpark, XGBoost, LightGBM, Scikit-learn, Streamlit

April 2025

- Built a scalable PySpark-based ML data pipeline over 73K USGS seismic records for feature engineering, preprocessing, and model experimentation.
- Engineered a custom Reliability feature from network station count, azimuthal gap and distance metrics, and applied polynomial expansion to linear models to capture non-linear seismic relationships.
- Trained and benchmarked 7 models across regression and classification tasks with the best model reaching RMSE of 0.029 and R^2 of 0.994, selecting Random Forest over GBT for deployment based on interpretability.
- Deployed a live Streamlit app at quakeanalysis.streamlit.app with a zip code risk checker, interactive heatmap and magnitude predictor.

Personal Engineering Portfolio Website | React.js, JavaScript, HTML5, CSS3

May 2026

- Designed and developed a responsive personal portfolio website using React and modern JavaScript to showcase professional experience, technical projects and engineering capabilities.
- Architected reusable React components and modular UI sections for navigation, project showcases, technical skills and contact interactions.
- Implemented a custom frontend design using HTML5, CSS3, responsive layouts and component-based development.
- Optimized usability, accessibility and responsive behavior across desktop, tablet and mobile environments.

Image Classification with Neural Networks | Python, TensorFlow, Scikit-learn

June 2025

- Designed a 10 layer CNN on Fashion MNIST (70K images, 10 classes) with a 64 to 32 to 16 filter progression and chose L2 regularization over dropout after observing dropout hurt convergence on this architecture, reaching 92.29% test accuracy.
- Built a feed-forward classifier on the UCI Adult dataset (30K records) using stratified splits to handle class imbalance and ColumnTransformer for mixed feature types, achieving 86% accuracy and macro F1 of 0.81.

PROFESSIONAL EXPERIENCE

Tata Consultancy Services

April 2024 – December 2024

System Engineer – Java Backend

Chennai, India

- Developed and shipped backend features for a high-traffic retail CRM serving a major Australian supermarket chain, contributing to 10+ Spring Boot microservices across the full software development lifecycle in an Agile team.
- Built and extended billing and transaction RESTful APIs handling customer order data across a distributed microservices architecture, following REST design principles and JSON data standards.
- Diagnosed and resolved database timeout errors on high-volume endpoints by identifying unoptimized queries and applying indexing fixes, improving application performance and response reliability on critical billing flows.
- Expanded JUnit test coverage across service classes, supported production and non-production environment troubleshooting and contributed to Docker and Kubernetes based deployments through Azure CI/CD pipelines on a bi-weekly release cycle.

Tata Consultancy Services

January 2023 – March 2024

Assistant System Engineer – Data Integration

Chennai, India

- Built and maintained enterprise ETL pipelines using Informatica PowerCenter and IICS integrating data across SAP, Oracle, GCP, Salesforce and Kronos HRM.
- Debugged and refactored existing pipelines by fixing transformation logic and validating outputs before each production release.
- Traced and resolved job failures from server issues and connection timeouts on scheduled batch pipelines, restoring delivery within SLA windows.
- Automated data validation using SQL and shell scripts, eliminating manual checks on routine pipeline runs.

Tata Consultancy Services

December 2021 – December 2022

Assistant System Engineer – Trainee

Chennai, India

- Completed onboarding in ETL development, SQL, Linux and Informatica and moved into production pipeline support within 3 months of joining.
- Contributed to production workflows through data validation, issue triage and pipeline fixes on live deployments.

AWARDS

- Star of the Month (2x), Tata Consultancy Services, for outstanding performance and delivery contributions.
- On-the-Spot Award, Tata Consultancy Services, for delivering a production change with zero business disruption.